

Outliers, Not Bits: Integer-Only Edge Inference for a Native 1-bit LLM

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Abstract

Native 1-bit LLMs such as BitNet b1.58 are trained with ternary weights and 8-bit activations (W1.58A8) and are built for *on-device*, *CPU/edge* inference, where there is no GPU, memory and energy dominate, and an integer-only datapath is the prize. We ask a sharper question than “does it run in integers”: running a *fully*-integer forward (integer-exact ternary matmuls, integer RMSNorm, RoPE, softmax, KV-cache, and integer multiply-shift rescaling) on the released BitNet b1.58 2B4T weights, *which components actually limit fidelity, and why?* Against a faithful float oracle on a held-out corpus (with a WikiText-2 slice and bootstrap confidence intervals) we find a consistent inversion of the usual intuition. The parts assumed to be delicate are nearly free: a **16-entry** exponential lookup table and 4-bit probability accumulation leave greedy generation unchanged, and the rescale reduces to an integer multiply and a shift with no loss. The parts that are delicate are about *where the quantization grid sits*, not how many bits it spends. (i) For the KV-cache, a clean granularity $\times$ symmetry ablation shows per-channel granularity is necessary and a zero-point helps only on top of it —per-vector *asymmetric* is *worse* than per-vector symmetric—and 2-bit per-channel-asymmetric KV is nearly free (+10 PPL on WikiText-2 vs. +266 for per-vector), traceable to a measured per-channel bias in the Key cache that is $2.5\times$ that of the Value cache. (ii) The single fragile knob is the activation scale: outlier-robust (mean/median) scaling raises perplexity by 3–6 orders of magnitude, because the int8 activations are heavy-tailed and the outliers carry the signal. (iii) Acting on (ii), we characterize the model’s post-hoc activation frontier: the established rotation + asymmetric-per-group recipe (no retraining, ternary weights untouched) takes activations to **int4** (free), **int3** ($\approx +2$ PPL on WikiText-2), and **int2** ($\approx +5$ PPL, less on easier text), where naive quantization collapses by 3–6 orders of magnitude; **int1** is the floor. The recipe is not new (rotation is QuaRot’s; concurrent PTQ methods TWLA/BWLA reach the W1.58A4 regime by re-quantizing weights, which we do not)—our angle is the frontier of the *shipped* model and the principle behind it: rotation *preserves* the load-bearing outlier information (an invertible basis change) whereas robust scaling *destroys* it—and because 2-bit, like the KV cache, needs an asymmetric (zero-point) code, not a symmetric one. Every finding replicates on a second, architecturally distinct 1-bit model (Falcon3-1B-1.58, SwiGLU with an untied head, vs. BitNet’s squared-ReLU with subln): same KV-granularity ordering, same per-channel-asymmetric 2-bit rescue, same activation-scale collapse, and the same measured Key per-channel bias (0.79σ in both). We also document the undocumented I2.S packed-ternary format, reverse-engineered to make the study possible. Code, a verified reference and Rust port, and a one-command harness to extend to full corpora and other model sizes are released.

1 Introduction

Native 1-bit LLMs [1, 2] are a bet on the edge. Ternary weights $\{-1, 0, +1\}$ turn matrix multiplication into additions and 8-bit activations keep the datapath integer, so the target is CPU and on-device hardware—phones, microcontrollers, integer NPUs—where there is no GPU to lean on and where memory footprint, memory bandwidth, and energy, not FLOPs, set the limit. The reference engine bitnet.cpp [4] is a CPU engine, and the BitNet b1.58 2B4T [3] model card explicitly disclaims GPU efficiency gains through the standard stack. On such hardware the relevant question is not “does it run in integers” but *which integer operations cost the most and which can be made cheap*: an *end-to-end integer* forward also does normalization, rotary embeddings, attention, the KV-cache, and the per-tensor/per-token rescaling, and it is not obvious which of these limits fidelity when pushed, nor where on-device memory should be spent (KV-cache bits, activation bits).

We answer this for the released weights. We build a verified reference from scratch (the GGUF uses an undocumented packed ternary format, `I2.S`, that we reverse-engineer in Section 3), establish a float oracle that generates coherent text, and run a fully-integer forward against it as a controlled experiment: every integer approximation is a knob, scored by perplexity on a held-out corpus with bootstrap confidence intervals.

Contributions. We are deliberately precise about what is new, because two of our phenomena are known for full-precision models and our value is in the 1-bit setting and in the decomposition, not in rediscovery.

1. **The non-matmul integer ops are nearly free, quantified** (Section 5). The ternary $\times$ int8 matmul is exact by construction; the contribution is showing that *everything around it*—integer RMSNorm, RoPE, a 16-entry exp LUT softmax, 4-bit probability accumulation, and an integer multiply-shift rescale—is also nearly free. This is not obvious a priori and is directly actionable: integer attention needs no costly exponential unit.
2. **A granularity $\times$ symmetry decomposition of KV quantization with confidence intervals** (Section 6). The corpus-robust result, holding on both a mini-corpus and a 10k-token WikiText-2 set, is that 2-bit KV is rescued only by per-channel granularity *and* a zero-point together—either alone is insufficient (on WikiText-2, per-channel-symmetric and per-vector-asymmetric both fail, +264 and +125, while per-channel-asymmetric is +10). *Which* single factor helps in isolation is itself corpus-dependent, which we report rather than hide. We tie the rescue to a measured per-channel Key bias, the native-1-bit instance of the structure exploited by KIVI [13] and KVQuant [14].
3. **The post-hoc activation frontier of the *shipped* model, and one principle** (Section 8). Leaving the trained ternary weights untouched, the established rotation + asymmetric-per-group recipe takes the shipped model’s activations to **int4** (free), **int3** ($\approx +2$ PPL), and **int2** ($\approx +5$ PPL on WikiText-2); int1 is the floor. We do *not* claim this recipe is new: rotation for low-bit activations is QuaRot’s idea [9], and concurrent PTQ methods reach the W1.58A4 regime with related rotations (TWLA [11], BWLA [12])—though by re-quantizing weights from full precision, which we do not. Our contribution here is the *characterization* of where the frontier sits on the already-shipped W1.58A8 model, plus the principle that organizes it and finding (ii): outlier *information* must be preserved (rotation moves it, robust scaling deletes it), and granularity-plus-zero-point is one lever against two different outlier structures.
4. **The I2.S format and a verified, reproducible artifact** (Section 3): the exact packing,

plus a numpy reference and a Rust port checked bit-for-bit, and methodological cautions (greedy-decoding collapse and logit cosine both mislead).

2 Background and related work

Native 1-bit LLMs. BitNet [1] introduced BitLinear; BitNet b1.58 [2] fixed weights to $\{-1, 0, +1\}$ with per-tensor absmean weight scaling and per-token absmax int8 activations (W1.58A8); the 2B4T model [3] is the first open release at scale, with the bitnet.cpp engine [4]. BitNet v2 [5] *re-trains* with an online Hadamard transform to smooth activation outliers and reach 4-bit activations; our Section 8 obtains 4-bit activations on the already-shipped b1.58 model without retraining.

Activation outliers and rotations. Outliers obstruct low-bit inference: LLM.int8() [6] keeps outlier channels in higher precision, SmoothQuant [7] migrates scale into weights, and AWQ [8] protects salient channels. Rotation-based methods (QuaRot [9]) multiply activations by an orthonormal (Hadamard) matrix to spread outliers before quantization; we apply exactly this idea, post-hoc, to a native-1-bit model whose weights we leave untouched. A complementary algorithm–architecture line treats outliers as a hardware problem: MicroScopiQ [10] keeps outliers at higher precision and prunes a fraction of low-importance weights to “fund” the extra outlier bits on a custom multi-precision accelerator—reaching the same conclusion from the systems side that outlier *handling*, not raw bit-width, is the design axis.

Concurrent work on PTQ for ternary-weight LLMs. A concurrent line of work from one group (Zhao et al.)—BWL A [12] (May 2026) and TWLA [11] (ICML 2026, June 2026)—uses rotation-based PTQ to reach W1AX / W1.58A4: ternary (or 1-bit) weights with low-bit activations. These are quantization *methods*: they ternarize/quantize weights from a full-precision model and optimize for accuracy at a target activation bit-width (TWLA contributes an asymmetric ternary *weight* quantizer, a Kronecker rotation that reshapes weights into ternary-friendly distributions, and inter-layer mixed precision, targeting A4). We are not proposing a quantizer. We leave the *already-trained* ternary weights of the shipped BitNet b1.58 2B4T untouched and instead *characterize* how far its activations can be pushed post-hoc with the (QuaRot-established) rotation + asymmetric-grouping recipe, map the int8→int1 frontier, and—disjoint from that program—decompose the KV-cache, show the non-matmul integer ops are free, replicate across architectures, and measure the on-device packing inversion. Their results show the low-bit-activation regime is reachable by weight re-quantization, a route disjoint from ours; our contribution is the characterization of the shipped model, the principle, and the systems result.

KV-cache quantization. For full-precision LLMs, KIVI [13] achieves 2-bit KV with per-channel (Key) asymmetric quantization because Keys have channel-wise outliers; KVQuant [14] quantizes Keys per-channel before RoPE; RotateKV [15] uses rotations. Our KV section reproduces the per-channel-asymmetric insight in the 1-bit regime and adds the granularity×symmetry decomposition with CIs and a pre-RoPE test.

3 The released model and its I2_S format

BitNet b1.58 2B4T has 30 layers, model dimension 2560, FFN width 6912, grouped-query attention (20 query / 5 KV heads, head dimension 128), RoPE base 5×10^5 , RMS-norm $\varepsilon = 10^{-5}$, vocabulary

Table 1: Reference registry. The reference configuration is identical throughout (native int8 activations, float KV); only corpus/slice (and, for the short corpus, whether a BOS token is prepended) vary. Every Δ PPL in the paper is against the matching-slice reference.

Used in	Corpus	Tokens	Ref PPL
KV sweep (Tab. 2)	WikiText-2 test	10,240	27.4
Activation rotation (§8)	WikiText-2 test	512	22.7
KV ablation, mini-corpus column (Tab. 3)	mini-corpus (16 psg.)	387	12.2
Sub-int8, recovery (Tab. 4, Fig. 3)	short (6 psg., +BOS)	133	8.27
Granularity isolation (Tab. 5)	short (6 psg., no BOS)	133	5.58
Cross-model, Falcon3-1B (Tab. 6)	WikiText-2 test	1,020	64.2

128,256, tied F16 embeddings, and a squared-ReLU FFN with sub-layer norms before the output and down projections.

The GGUF stores ternary tensors in a packed type, `I2.S` (ggml type 36), not documented sufficiently to reproduce. We reverse-engineered it by requiring coherent generation. Each tensor is $\lceil n/4 \rceil$ bytes of 2-bit codes, then one little-endian f32 scale, padded to 32 bytes. Crucially the codes are *128-element block interleaved*: within each 32-byte (128-element) block, for byte $j \in [0, 32)$ and 2-bit slot $s \in [0, 4)$ (LSB-first), the element index is $(3 - s) \cdot 32 + j$, and the value is $\text{code} - 1$. Sequential unpacking, or the non-reversed group order, yields the correct ternary *distribution* but scrambled positions and thus fluent gibberish—a trap worth flagging for re-implementers. The dequantized weight is $(\text{code} - 1) \cdot \text{scale}$.

4 Method

Oracle and integer path. The oracle dequantizes ternary weights, applies the model’s own per-token int8 absmax activation quantization (part of the trained model), and computes the rest in float64; it generates coherent text. The integer path mirrors it in integers: int8 activations, integer-exact int32 ternary accumulation, data-adaptive fixed-point RMSNorm with integer reciprocal-sqrt, Q15 RoPE, a fixed-point exp-LUT softmax, and—closing the last float operation—the per-(tensor,token) rescale as an integer multiply by a 24-bit mantissa and a shift. This fully-integer forward is token-exact with the oracle (logit cosine 0.9992; identical greedy text).

Metrics and corpus. We score per-token NLL / perplexity on a 16-passage multi-genre mini-corpus (387 tokens) and a real WikiText-2 [16] test set (10,240 tokens), with 95% CIs by paired token-level bootstrap (2000 resamples). For the KV and activation sweeps we keep RMSNorm/RoPE/softmax in float (justified by Section 5, which shows their integer forms are token-exact) so the perplexity deltas are attributable to the studied knob.

On the reference perplexity. Reference PPLs differ across tables because the *corpus and slice* differ, *not* the configuration: every reference is the model’s native path—absmax int8 activations and a float (unquantized) KV cache—and every reported Δ PPL is computed against the reference on the *same* slice, so all deltas are within-slice and apples-to-apples. Table 1 is the registry of references; e.g. the two WikiText-2 numbers (27.4 and 22.7) are the same configuration on a 10,240- vs a 512-token slice, not an easier baseline for any result.

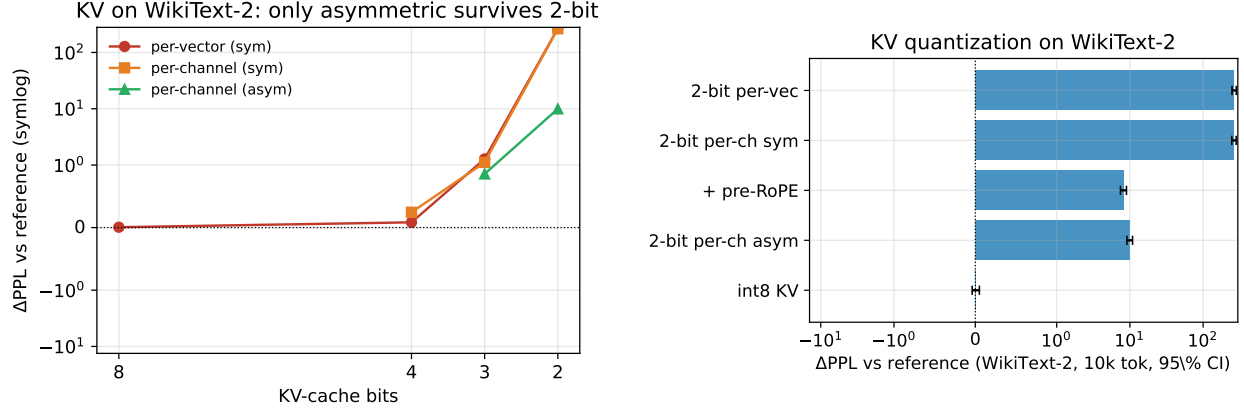


Figure 1: KV-cache quantization, both panels on WikiText-2 (10,240 tokens). Left: Δ PPL vs bit-width by granularity (symlog). Right: the 2-bit configs with bootstrap CIs. Per-vector *and* per-channel-symmetric 2-bit both collapse (+264–266); only per-channel-*asymmetric* keeps 2-bit nearly free (+10). At this corpus size the zero-point, not granularity alone, sets the cliff.

5 The non-matmul integer ops are nearly free

The ternary $\times$ int8 product is exactly int32-representable, so the matmul itself is lossless; the question is everything else. At int8 activations and int8 KV, the fully-integer forward matches the oracle token-for-token (final-layer cosine 0.9997, per-layer cosine ≥ 0.998 across all 30 layers). Integer softmax is strikingly cheap: a **16-entry** exp LUT still yields identical greedy text (logit cosine 0.993) and 4-bit probability accumulation is harmless (logit cosine 0.977); the integer multiply-shift rescale loses nothing (0.9992). *The expensive parts of integer attention—the exponential and the normalization—turn out not to be expensive.* The interesting failure modes therefore lie elsewhere.

6 The KV cache: granularity, not bit-width

Figure 1 sweeps KV bit-width for three granularities. Per-vector KV degrades gracefully through 3 bits and collapses at 2 bits; the 3-bit “collapse” seen under single-prompt greedy decoding is a *decoding* artifact (a flipped argmax)—in perplexity, per-vector 3-bit costs only +1.3 on WikiText-2 (Table 2). (Where a quantized configuration shows a small *negative* Δ PPL—e.g. int8 KV at -0.0 to -0.2 in Tables 2, 6—it is quantization acting as mild regularization at the third significant figure, well within the bootstrap CIs, not a real improvement.)

The ablation (Table 3) decomposes the 2-bit rescue. The corpus-robust conclusion is that *both* per-channel granularity *and* a zero-point are required: their combination (per-channel-asymmetric) is the only config that holds at 2-bit on both corpora (+0.9 on the mini-corpus, +10 on WikiText-2). Which single factor helps *in isolation* is corpus-dependent and we flag it rather than over-read it: on the mini-corpus per-channel-symmetric already helps (+11) while per-vector-asymmetric hurts (+175, a zero-point wasting levels when a vector spans an outlier); on the larger WikiText-2 set this reverses (per-channel-symmetric collapses to +264, per-vector-asymmetric helps at +125). The robust signal is the conjunction, not either marginal.

Mechanism. Figure 2(left) measures the per-channel bias of the Key and Value caches, $|\mu_c|/\sigma_c$. Key channels are off-center by $\approx 0.79\sigma$, $2.5\times$ the Value cache (0.32σ): a symmetric quantizer,

Table 2: Integer-path KV-cache perplexity on WikiText-2 test (10,240 tokens). Int8 KV is free (CI spans 0); 2-bit *per-vector and per-channel-symmetric both collapse* (+264–266), and only the *asymmetric* per-channel code rescues 2-bit (+10)—so at this corpus size granularity alone is not enough, the zero-point is decisive. Robust activation scaling is catastrophic.

Configuration	PPL	Δ PPL	95% CI on Δ PPL
reference (absmax, KV float)	27.4	+0.00	[+0.00, +0.00]
per-vector 8-bit	27.4	+0.01	[−0.04, +0.05]
per-vector 2-bit (sym)	293.2	+265.7	[+247.9, +283.7]
per-channel 2-bit (sym)	291.5	+264.0	[+246.2, +282.7]
per-channel 2-bit (asym)	37.4	+9.96	[+9.11, +10.87]
+ pre-RoPE Keys	35.6	+8.17	[+7.43, +8.98]
absmean activation scale	1.7e5	+171080	[+157977, +185339]

10,240 tokens; paired token-level bootstrap, 2000 resamples.

Table 3: 2-bit KV-cache ablation, Δ PPL on two corpora. The robust result is the conjunction: only per-channel *and* asymmetric holds on both. Which factor helps *alone* flips with corpus size—per-channel-symmetric helps on the mini-corpus but collapses on WikiText-2; per-vector-asymmetric hurts on the mini-corpus but helps on WikiText-2.

2-bit KV configuration	mini-corpus (387 tok)	WikiText-2 (10k)
per-vector, symmetric	+86.9	+265.7
per-vector, asymmetric	+175.1	+124.6
per-channel, symmetric	+11.0	+264.0
per-channel, asymmetric	+0.9	+10.0

paired token-level bootstrap, 2000 resamples.

centered at zero, wastes about half its range on a biased channel, so a per-channel zero-point recovers roughly a bit. This is the native-1-bit instance of the Key-channel structure exploited by KIVI [13] and KVQuant [14]; quantizing Keys pre-RoPE (Table 2) is marginally better, consistent with RoPE mixing channels.

7 Activation scaling is the load-bearing knob

The one component with no slack is the activation scale. Replacing absmax with outlier-robust alternatives is catastrophic: on WikiText-2, absmean reaches 1.7×10^5 perplexity (reference 27.4). Figure 3 sweeps the absmean clip multiplier: perplexity falls only as the multiplier grows past the outliers—i.e. as the “robust” scale stops being robust. The cause is structural: the activation absmax/absmean ratio has median 7.8, p90 $\approx$ 106, max $\approx$ 1460 (Figure 2, right). In a quantization-aware-trained W1.58A8 model the outliers *are* the signal.

8 Preserve, don’t suppress: the post-hoc activation frontier

Section 7 and this one look contradictory: there, touching the activation outliers is catastrophic; here, we transform them and low-bit quantization becomes nearly free. The resolution is the central principle of this paper, and it is sharper than “outlier handling is the design axis”: **outlier information must be preserved; what is negotiable is only whether it is concentrated**

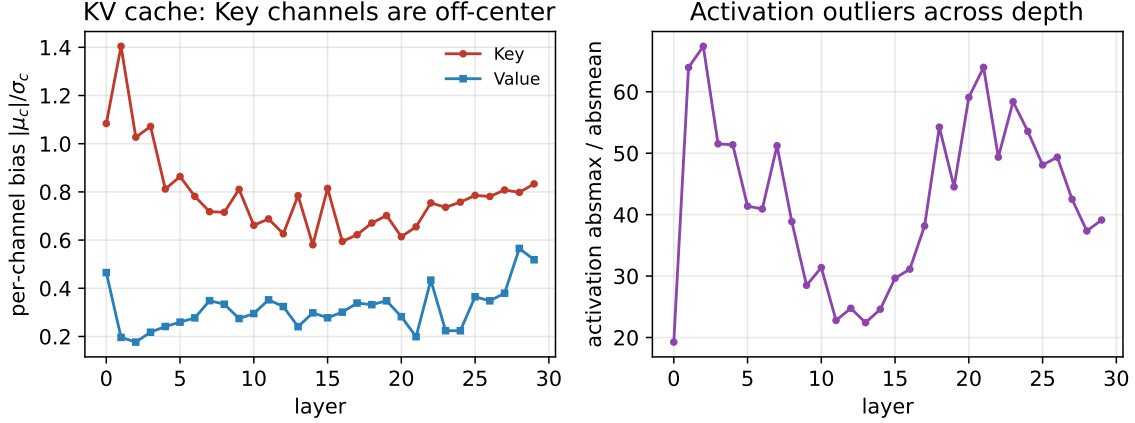


Figure 2: Mechanism. Left: per-channel bias $|\mu_c|/\sigma_c$ of Key vs Value across layers—Keys are systematically off-center, explaining why asymmetric per-channel KV recovers 2-bit. Right: activation absmax/absmean across layers (mean 42.6), the heavy tails behind the activation-scale fragility.

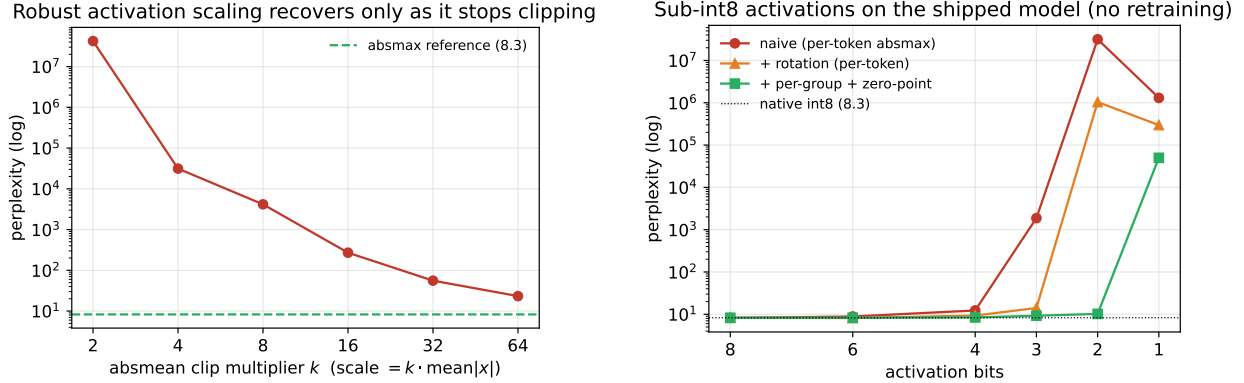


Figure 3: Left: robust activation scaling recovers only as it stops clipping outliers. Right: post-hoc activation quantization of the shipped model—naive collapses below 8 bits; rotation reaches 4-bit; rotation + per-group scaling + a 2-bit zero-point (the KV cache’s recipe) reaches **2-bit**, with 1-bit the floor (Section 8).

or spread. Robust (mean/median) scaling is *information-destroying*—it clips the outlier values and discards them. An orthonormal rotation is *information-preserving*—an invertible change of basis that redistributes the same energy across coordinates without losing any. The two results that appear to conflict are one law shown twice: one operation deletes the signal, the other merely moves it.

This sharpens the novelty relative to QuaRot [9], which rotates a full-precision model to remove outliers it treats as a *nuisance*. Here Section 7 has just shown the outliers are load-bearing *signal*, so the a-priori prediction is that rotating them away destroys the model. It does not—precisely because rotation preserves rather than suppresses—and that non-obviousness is the contribution, not the port.

Concretely we quantize activations as $x \mapsto R^\top Q_b(Rx)$ with $R = H_p \otimes O_m$ a normalized Sylvester Hadamard on the largest power-of-two factor of the activation dimension and a fixed small orthogonal block on the remainder ($O(d \log d)$ cost); the ternary weights and the model are untouched and

nothing is retrained. Per-token rotation already recovers 4-bit activations (+0.4 PPL, Table 4), reaching *post-hoc* the int4-activation regime BitNet v2 [5] reached by *retraining*. We are careful about what this claims. A same-model head-to-head is not available: BitNet v2 reports its int4 results on separately retrained 1.3B/3B/7B models, and no int4-activation weights are released for the 2B4T checkpoint we use, so there is no shared (model, benchmark) pair to compare on. Both reach near-lossless int4 (theirs by retraining, ours post-hoc at +0.14 PPL), but on different checkpoints; and the usual PTQ-vs-QAT prior says retraining should still win at matched bits. We therefore claim the same bit-width *regime*, reached without retraining—not demonstrably the same quality as a retrained model.

Granularity and a zero-point are the same lever for activations as for the KV cache.

The KV-cache rescue (Section 6) came from finer granularity *and* a zero-point; the identical pair applies here. Adding a per-group(32) scale on top of the rotation extends the recovery to 3 bits (+3.5 $\rightarrow$ +0.45 PPL on the short corpus, Table 5); adding a per-group *zero-point* (asymmetric quantization) then reaches **2 bits**, where a symmetric code fails. The decisive factor is exactly the one the KV cache flagged: a symmetric “2-bit” code wastes a level (it is effectively ternary), while an asymmetric 4-level code uses all four. We report the magnitudes on a harder corpus and are explicit that the int2 cost is corpus-dependent. On a WikiText-2 slice (512 tokens, reference 22.66; the 10,240-token KV set of Table 2 has reference 27.4—same configuration, longer slice, Table 1): **int4** is free (22.54, -0.1), **int3** is +2.1, and **int2** (per-group(8) asymmetric + rotation) is +5.3 PPL—usable, and far below the +956 of symmetric int2 or the $\sim 10^6$ of naive int2 on the *same* slice, but a real cost, not the +0.7 the 133-token corpus suggested. So int2 activations are reachable post-hoc by the asymmetric-granular recipe (the zero-point is the load-bearing piece), at a corpus-dependent cost of roughly 1–5 PPL.

Where post-hoc activation quantization stops: 1 bit. To be explicit about *which* bit-width: this entire section varies the *activation* precision while the ternary 1.58-bit *weights* are untouched (the model is W1.58A8; “1-bit” in BitNet refers to weights, which we never change). For activations, the post-hoc floor is one bit, not two: with only two levels, even fine-group asymmetric quantization, rotation, and keeping the 128 largest channels per token at int8 leaves 1-bit activations at $\sim 2 \times 10^4$ PPL ($\sim 4000\times$ the reference). This is the expected side of a well-known asymmetry—static weights tolerate far lower bit-widths than dynamic, outlier-laden activations, and sub-4-bit activations are known to require QAT [5]; we make no claim that 1-bit activations are unreachable *with* retraining, only that no post-hoc method we tried reaches them. The honest post-hoc reach is **int8 $\rightarrow$ int2 activations** on the shipped model (int4 free, int3 $\approx$ +2 PPL, int2 $\approx$ +1–5 PPL corpus-dependent), with 1-bit the post-hoc floor—and 2-bit is reached by the *same* asymmetric-granular recipe the KV cache required.

9 Cross-architecture generalization

To test whether these are properties of one model or of native-1-bit inference, we repeat the headline experiments on Falcon3-1B-1.58, an architecturally distinct 1-bit model (Llama-style: SwiGLU FFN, no sub-layer norms, an untied LM head, 18 layers, head dim 256), using the identical harness (the architecture profile is auto-detected from the GGUF). Table 6 reports Δ PPL *on WikiText-2* (the same corpus as our main KV evidence, not the mini-corpus). Every qualitative result transfers: int8 KV is free; at 2 bits per-vector *and* per-channel-symmetric KV both collapse (+105 and +108 for Falcon3) while only per-channel-*asymmetric* rescues it (+18.5); robust activation scaling is

Table 4: Sub-int8 activations on the shipped b1.58 (no retraining), perplexity (native int8 reference 8.27). Per-token rotation alone collapses below 4 bits; the bottom row adds the KV-cache recipe (fine per-group scaling and, at 2 bits, a zero-point), which reaches 2-bit. 1-bit is the floor.

activation bits	8	6	4	3	2	1
naive (per-token absmax)	8.18	8.86	12.31	1877	3.2×10^7	1.3×10^6
+ rotation (per-token)	8.27	8.15	9.31	14.09	1.0×10^6	3.0×10^5
+ per-group, +zero-point at 2b	—	—	8.41	9.24	10.21	$\sim 5 \times 10^4$

Bottom row: rotation + per-group(32) at 4/3-bit, + asymmetric per-group(8) at 2-bit; native-int8 reference 8.27. 1-bit resists every method.

Table 5: The granularity + zero-point recipe, isolated (BitNet, short corpus, ref 5.58): Δ PPL at 2-bit. Per-group scaling alone is not enough; the *zero-point* (asymmetric, 4-level vs. symmetric 3-level) is what makes 2-bit work—the same factor that rescued the 2-bit KV cache. The mechanism holds on the larger WikiText-2 slice: there per-group(8) asymmetric int2 is +5.3 PPL while symmetric is +956 (Section 8); the asymmetric *rescue* is corpus-robust even though the absolute int2 cost grows on harder text.

2-bit activation quantizer (with rotation)	Δ PPL
per-token, symmetric	$\sim 10^6$
per-group(32), symmetric	+1460
per-group(16), asymmetric	+1.4
per-group(8), asymmetric	+0.69

catastrophic; and the mechanism is quantitatively the same—the Key per-channel bias is 0.79σ in *both* models. Two architectures at different scales (2B and 1B), built by different groups, exhibit the same integer-inference failure surface on the same corpus, which is the evidence that this is about native-1-bit *quantization-aware training*, not a quirk of one checkpoint.

We flag the Key-bias coincidence rather than hide it: the per-channel bias matching at 0.79σ to two significant figures across two architectures is more than our ~ 1.5 k-token estimate resolves, so we read it as “both clearly elevated and near each other,” not as a measured constant. We conjecture it is a genuine attractor—absmax activation quantization plus RoPE applied to the Key projection should induce a similar per-channel offset regardless of width or FFN type—but a larger corpus and more models are needed to decide between “invariant” and “coincidence.” It is exactly the kind of striking regularity that should be stated as a prediction (Section 10), not oversold as a law.

10 Discussion, implications, and predictions

The integer-inference picture for this native-1-bit model inverts the usual intuition. The fragile components are not the arithmetic primitives (integer attention, fixed-point exp, softmax normalization, KV bit-width to 3 bits) but *where the quantization grid sits and how fine it is*. The unifying law, stated once: outlier *information* must be preserved, and the only free variables are how finely the grid is placed (granularity) and whether the basis concentrates or spreads that information. Robust scaling loses; rotation and per-group/per-channel granularity win—for the KV cache and for activations alike. To be precise about “same lever”: granularity localizes the quantization grid in both cases, but the *outlier structure* it localizes against differs—the KV cache has per-channel

Table 6: Cross-architecture generalization *on WikiText-2* (Δ PPL vs each model’s reference). The same failure surface holds across two architectures built by different groups: int8 KV is free; 2-bit per-vector *and* per-channel-symmetric KV both collapse; only per-channel-*asymmetric* rescues 2-bit; robust activation scaling is catastrophic; and the Key per-channel bias is the same 0.79σ in both. (BitNet on a 10k-token set, Falcon3-1B on a 1k-token set; see Table 1.)

	BitNet b1.58 2B4T	Falcon3-1B-1.58
2-bit KV / activation config	(ref 27.4)	(ref 64.2)
int8 KV (per-vector)	+0.01	−0.27
2-bit KV, per-vector	+265.7	+104.5
2-bit KV, per-channel (sym)	+264.0	+107.7
2-bit KV, per-channel (asym)	+ 10.0	+ 18.5
absmean activation scale	$+1.7 \times 10^5$	$+4.2 \times 10^3$
Key per-channel bias $ \mu_c /\sigma_c$	0.79	0.79

bias (fixed by a zero-point), activations have heavy magnitude *tails* (fixed by a rotation). The shared principle is preserve-and-localize; the local remedy is matched to the local structure.

Implications for edge inference (memory exactly; latency/energy expected). Every result is a lever for on-device deployment, where memory and bandwidth—not FLOPs—bind. We quantify the memory effects, which are exact functions of the model dimensions; latency and energy are *expected* consequences we do not measure on hardware here (Section 12). (1) An integer-only datapath is achievable end-to-end, including the rescale (integer multiply-shift), so no floating-point unit is required, and a *16-entry* exp LUT suffices for attention, so no precise integer exponential is needed—both matter on microcontrollers and integer NPUs. (2) KV cache: for BitNet 2B (30 layers, 5 KV heads, head dim 128) the cache is 37.5 KB/token at int8, so a 4096-token context costs ≈ 150 MB; per-channel-asymmetric **2-bit** KV cuts this to 9.4 KB/token and ≈ 38 MB ($4\times$) for < 10 PPL—more on-device context per byte of SRAM/DRAM. (3) Activations: a free $O(d \log d)$ rotation plus the same fine-group asymmetric quantization reaches **int2** with no retraining (Section 8), roughly a $4\times$ cut in activation memory and bandwidth versus native int8. The headline: on the edge the bits to cut are in the KV-cache and activations (via grid placement and granularity), not in the softmax or attention precision.

Predictions (falsifiable, testable with our harness). (1) The Key per-channel bias, and thus the per-channel-asymmetric KV rescue, will appear in any RoPE-based 1-bit model; (2) the activation absmax/absmean ratio predicts how badly robust scaling fails and how few bits rotation buys; (3) applying the rotation *before* RoPE to Keys, and combining it with per-channel-asymmetric KV, will compound (rotation reduces channel bias, asymmetry absorbs the residue). Our single-model results are the first data point; the released harness is built to test these across models and a full corpus.

11 Hardware measurements (aarch64)

BitNet’s target is ARM CPUs, and on such hardware decoding is *memory-bound*: each token streams the weights and the KV cache, so bandwidth, not FLOPs, sets the latency. We measure this directly with a small C microbenchmark on an aarch64 CPU (NEON, int8 dot-product /

i8mm) and report it in Table 7. The sustained read bandwidth is 47.5 GB/s; the 2-bit-packed ternary weights are 0.52 GB, so the weight-streaming latency floor is 11 ms/token (≈ 90 tok/s, single-thread), in line with the 29 ms/token end-to-end that bitnet.cpp reports for this model [3]. The integer arithmetic is *not* the bottleneck (it hides under the memory stream with int8 SIMD), which is exactly why an integer-only datapath is the right edge target. The KV-cache result is now *measured*, not inferred: at a 4096-token context the cache is 157 MB at int8 and 39 MB at 2-bit, and the corresponding per-token KV memory traffic falls from 3.3 to 0.8 ms—so the per-channel-asymmetric 2-bit KV cache of Section 6 buys a $4\times$ reduction in both footprint and bandwidth on real silicon. End-to-end, our pure-Rust port (Section 13) decodes the real model on an Apple-silicon laptop, and the two weight-storage choices expose the memory–compute boundary directly (Table 7, bottom). With weights *unpacked* to int8 it runs at 12.8 tok/s but is *memory-stalled* (IPC ≈ 1.0 , 4.2 GB). Storing them *2-bit packed* cuts the footprint to 2.7 GB as predicted, but decode drops to 6.1 tok/s: it is now *compute-bound*, at IPC ≈ 3.9 while executing $8\times$ more instructions to unpack the bits in scalar code. On this CPU the workload was not actually bandwidth-bound (~ 25 of ~ 100 GB/s), so packing trades memory stalls for unpacking compute.

The boundary condition (a prediction). This is not a one-device anecdote but an instance of a predictable threshold. Write decode time per token as a roofline $t_{\text{unpacked}} = \max(W/B, M/C)$ and $t_{\text{packed}} = \max(W/4B, M/C+U)$, where W is the weight bytes streamed, B memory bandwidth, M/C the matmul compute time, and U the per-token unpack cost. Packing improves latency *iff* the device is weight-bandwidth-bound (W/B is the binding term) *and* the unpack stays cheap enough, $U < W/B - W/4B = \frac{3}{4} W/B$. Our laptop sits on the wrong side—compute-bound, so packing loses—while a bandwidth-starved target (a phone, or a many-core part where C is large and B small) sits on the other, where packing wins. The corollary is that a cheap unpack (small U , i.e. the SIMD shuffle kernel bitnet.cpp uses—NEON unpacks 16 bytes to 64 ternary values in a few instructions) moves the threshold so packing wins almost everywhere. Mapping this boundary across devices (B/C ratios), bit-widths, and unpack implementations is a natural and, we think, sharper follow-up; here we locate it at one point and give the rule. (Both artifacts are one command: `bench/kernels.c` and `cargo run`.)

12 Limitations

We study two models (BitNet b1.58 2B4T and Falcon3-1B-1.58); the corpus is small (a 387-token mini-corpus and a 10,240-token WikiText-2 set) to fit a CPU/numpy reference, so absolute perplexities are not benchmark numbers, though the relative effects and their CIs are stable across two corpora *and* two architectures. The KV/activation sweeps keep norm/RoPE/softmax in float (the fully-integer forms are validated token-exact in Section 5, and the rescale is integer multiply-shift), and the rotation study is post-training fake-quant. On the systems side we measure memory bandwidth, weight/KV streaming latency, and an end-to-end decode of the Rust port on Apple silicon (Section 11); what we do *not* yet report is an *optimized* engine—the port’s scalar kernels are either memory-stalled (unpacked, 12.8 tok/s) or spend their cycles unpacking (2-bit packed, 6.1 tok/s), so both are floors, and an energy measurement is left to future work. Closing the gap with a SIMD shuffle-unpack kernel (to get the packed footprint at the unpacked speed) is engineering the characterization directly motivates, not a change of method. The remaining empirical breadth—full-corpus perplexity, the rotation result on the larger slice, and a size ladder (Falcon3-3B/7B)—is a cheap extension with the released one-command harness (BN_CORPUS, BITNET_GGUF), not a change of method.

Table 7: Hardware measurements, BitNet 2B. *Top*: microbenchmark (`bench/kernels.c`) on an aarch64 CPU—decoding is memory-bound, so the 2-bit weights/KV turn bandwidth into latency and footprint wins. *Bottom*: end-to-end decode of our pure-Rust port on an Apple-silicon laptop. The port stores weights *unpacked* (int8) for simplicity, so its footprint (4.2 GB) is 4× the 0.52 GB a packed kernel would use; its throughput is single-core-scalar parallel (no NEON), an honest floor.

Quantity (4096 ctx)	Measured
<i>microbenchmark (aarch64)</i>	
Sustained memory read bandwidth	47.5 GB/s
2-bit packed weights	0.52 GB
weight-stream latency floor	11 ms/token (≈ 90 tok/s)
KV cache footprint, int8 $\rightarrow$ 2-bit	157 $\rightarrow$ 39 MB (4×)
KV memory traffic/token, int8 $\rightarrow$ 2-bit	3.3 $\rightarrow$ 0.8 ms
<i>end-to-end Rust port (Apple silicon), threaded</i>	
unpacked int8 weights	12.8 tok/s, 4.2 GB
memory-stalled (IPC ≈ 1.0)	
2-bit packed weights	6.1 tok/s, 2.7 GB
compute-bound unpack (IPC ≈ 3.9)	

13 Reproducibility

We release a from-scratch reference (numpy oracle and integer engine), a pure-Rust port with the I2S decoder checked bit-for-bit against Python, the perplexity / bootstrap / mechanism / figure scripts, and the rotation study. Every number here is produced by a script in the repository. Code, the per-token NLL arrays backing every table and figure, and reproduction instructions are at <https://github.com/kksante/bitnet-int8-rs> (commit b7c9846); model weights are the public Microsoft and TII GGUF releases, not redistributed here.

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